Andres Pena

EMBEDDED SYSTEMS | AVIONICS | ELECTRONICS TEST

Olney, MD | (301) 633-8696 | andrfp@swort.net

andres.swort.net | linkedin.com/in/andrespenaavila

Embedded systems technical lead with paid avionics development experience. Hands-on C/C++, Python, PCB review, prototype assembly, soldering, hardware troubleshooting, and validation.

Engineering Experience and Leadership

SWORT Aerospace | Technical Lead | 2023–Present

- Lead paid flight-controller programs, including 50 × 50 mm X8 avionics; own customer requirements, system architecture, schematic/PCB reviews, contractor coordination, and prototype acceptance planning.
- Direct prototype troubleshooting, corrective actions, design revisions, and repeat testing before release.
- Assemble, solder, and troubleshoot electronics with an oscilloscope, multimeter, and bench power supplies.
- Support projects under nondisclosure agreements; protect proprietary files and control agreed disclosures.

ASME / Shell Eco-marathon | Electromechanical Lead | 2025–Present

- Lead ongoing STM32 vehicle telemetry and motor-controller development: sensor data acquisition, offline logging, driver display, and CSIA shadow monitoring.
- Designed and integrated the vehicle's hydraulic brake system; created SolidWorks CAD and MATLAB simulations projecting ~12% shorter stopping distance.

Purdue Space Program / ASTRA | Avionics Engineer | 2025

- Developed C/C++ flight-control firmware and avionics sensor interfaces for GNC testing; performed LTspice circuit analysis, PCB layout review, and board-level fault isolation.

Selected Projects and Research

CSIA Sensor Assurance Research: Developed a 25 Hz Python/MAVLink sensor-diagnosis runtime; evaluated 26 public PX4 logs and live ArduPilot SITL tests. Recorded 277,500 desktop soak-test ticks with zero runtime errors. Integrated a zero-allocation native C reference core; verified C/Python parity to 1e-12 on an 8-node/11-edge graph in desktop tests.

STM32H743 Flight Controllers: Lead four-layer STM32H743 flight-controller prototypes with dual IMUs, barometers, and magnetometers; review schematics, layouts, BOMs, and manufacturing files. Validation ongoing.

Custom ESC Development: Lead architecture definition for eight-channel ESC systems: paired 4-in-1 boards, AM32, bidirectional DShot600, per-motor current/temperature telemetry, and power/thermal requirements. Build pending.

Hybrid EMI Filter Research: Developing an active/passive filter concept for electric-aircraft inverters, supported by Python stability models, a 1,088-case synthetic sensitivity screen, fault analysis, and a staged bench-verification plan. Hardware validation pending.

Code and validation documentation: github.com/SWORT-Aerospace

Technical Skills

Programming: C/C++, Python, MicroPython, MATLAB; Linux, Git, Docker, pytest

Embedded: STM32H7, ESP32, Raspberry Pi; UART, SPI, I2C, CAN, MAVLink; PX4/ArduPilot SITL

Electronics & CAD: PCB/schematic review; Altium, KiCad, EasyEDA Pro, LTspice, SolidWorks, Onshape

Test & Build: Soldering, assembly, hardware testing, fault injection, regression testing, bill of materials (BOM)

Manufacturing: Additive manufacturing, metal-part fabrication, chemical manufacturing

Education

Montgomery College | Electrical Engineering

Engineering studies: 2025–Present | Prior study: Purdue University

English and Spanish